

SHAKTI C-Class Dual-Issue Performance Model Experiments

perf-model

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Executive Summary

The current dual-issue SHAKTI model predicts 10,589,781 cycles and IPC 1.2060 on the generated dual CoreMark trace window. The clean dual RTL ground truth is 10,398,758 cycles and IPC 1.22817; the cycle-stamped run reproduced those app-log counters exactly, so the trace instrumentation is non-perturbing.

Validation gate status: not passed. The dual baseline now has a real per-instruction Δt comparison against RTL, not a proxy. Its accuracy is 93.302% (11,915,999/12,771,407 matching inter-commit deltas), with aggregate cycle error 1.837%. This is below the 97% Phase-1 gate, so the architectural experiments below are useful directional signals but still carry the measured residual timing error.

The backpressure blind spot that motivated this round is fixed for aggregate cycle deltas: the depth-1 s3/s4, s0/s1, s1/s2, and s4/s5 RTL slowdowns are now predicted within a few percent. However, local Δt cadence is still weak for the front-end depth-1 Dhrystone points, so this is not a full mechanism-level closure.

A single-issue floating-point validation addendum now closes the integer-only hole in the validation set. A new dense FP benchmark has 28.575% FLOAT instructions in its measured window. Before FP timing was modelled, the single-issue model reached only 64.612% Δt accuracy on that trace; after adding source-derived FPU latency it reaches 99.863%, with -0.063% aggregate cycle error.

The second memory port alone helps only modestly in this model: +0.698%. Intra-bundle ALU forwarding is the largest individual gain. The best draining two-issue combination found is E+H: intra forwarding + branch, with IPC 1.2953 (+7.406% versus baseline). The quad approximation did not complete/drain in the full-trace sweep and is reported as not completed, not as a measured null. The model-owned branch+branch opportunity count is much smaller than the defective RTL event-52 story suggests.

Experiment Summary

Experiment	Cycles	IPC	Δ IPC	Δ	Verdict
Baseline	10,589,781	1.2060	+0.0000	+0.000%	no effect
A: second memory port	10,516,356	1.2144	+0.0084	+0.698%	helped
B: decouple lockstep	10,517,329	1.2143	+0.0083	+0.689%	helped
C: A+B	10,483,424	1.2182	+0.0122	+1.015%	helped
D: independent retire	10,589,781	1.2060	+0.0000	+0.000%	no effect
E: intra-bundle ALU forwarding	9,968,077	1.2812	+0.0752	+6.237%	helped
F: relax branch next-PC stall	10,589,781	1.2060	+0.0000	+0.000%	no effect
G: symmetric slots	10,589,781	1.2060	+0.0000	+0.000%	no effect
H: second branch unit	10,539,244	1.2118	+0.0058	+0.480%	helped
I: best combination	10,011,897	1.2756	+0.0696	+5.772%	helped
J: quad issue on best	n/a	n/a	n/a	n/a	not completed

Additional Combination Probes

Experiment	Cycles	IPC	Δ IPC	Δ	Verdict
A+E: memory + intra forwarding	10,120,648	1.2619	+0.0559	+4.635%	helped
E+H: intra forwarding + branch	9,859,557	1.2953	+0.0893	+7.406%	helped

Validation

Item	Value
Dual Δt accuracy on generated CoreMark trace	93.302%
Dual Δt matches / compared	11,915,999 / 12,771,407
Dual RTL commit-span cycles from trace	10,398,784
Dual baseline model cycles	10,589,781
Dual clean RTL ground-truth cycles	10,398,758
Dual aggregate cycle error	1.837%
Dual baseline model IPC	1.20601
Dual clean RTL ground-truth IPC	1.22817
Held-out Dhrystone clean dual RTL IPC	1.33 aggregate only

Counter-profile validation. The model emits the RTL-facing counters needed for validation. The local cycle-stamped trace is CoreMark, while the supplied RTL counter profile is from an instrumented Dhrystone run. Absolute counter counts across those benchmarks are not a valid calibration target, so the mem+mem 5% counter gate remains open until either CoreMark RTL counters or a Dhrystone commit trace are produced.

The baseline CoreMark model counter profile is:

Counter	Model value
dual_issued	4,855,836 (45.854% of cycles)
raw_hazard	2,020,130
one_instr	149,723
st3_not_firing	2,022,438
mem_mem_hazard	729,996
mem LL / LS / SS	448,270 / 274,725 / 7,001
branch+branch opportunities	159,885
mispredict	217,257

For reference, the supplied Dhrystone RTL profile was: dual-issued about 50% of cycles, RAW 12,051, one-instruction fetch 4,026, stage3 not firing 4,864, mem+mem 22,510 (LL 6,010, LS 6,000, SS 10,499), and mispredict 2,034.

Held-Out Design-Point Validation

Benchmark	Baseline error	Baseline Δt	Overall floor	Small-effect floor
Dhrystone	-0.427%	99.655%	3.14 pp	0.17 pp
CoreMark	-0.790%	99.150%	1.46 pp	0.64 pp

Dhrystone keeps the original held-out baseline accuracy target. CoreMark baseline Δt is 99.150%, slightly below the requested 99.2% target, while aggregate cycle error remains within 1%. Variant CoreMark Δt is not reported because only the baseline CoreMark commit trace is archived; CoreMark variant rows are aggregate delta checks.

Dhrystone key backpressure points.	Point	Measured	Predicted	Error	Δt
	s3s4_1	+91.75%	+93.04%	+1.29 pp	98.40%
	s0s1_1	+75.68%	+74.50%	-1.19 pp	67.39%
	s1s2_1	+93.55%	+91.55%	-2.00 pp	68.01%
	s4s5_1	+90.19%	+90.92%	+0.72 pp	99.98%
	s3s4_2	+0.00%	+0.00%	-0.00 pp	99.65%
	s0s1_3	+0.01%	+0.00%	-0.01 pp	99.65%
	s1s2_3	-0.30%	+0.00%	+0.30 pp	99.34%
	Point	Measured	Predicted	Error	Δt
	s3s4_1	+77.62%	+77.97%	+0.35 pp	—
Coremark key backpressure points.	s0s1_1	+72.86%	+73.21%	+0.36 pp	—
	s1s2_1	+77.29%	+74.63%	-2.66 pp	—
	s4s5_1	+71.48%	+72.23%	+0.74 pp	—
	s3s4_2	+0.00%	+0.06%	+0.06 pp	—
	s0s1_3	+0.88%	-0.02%	-0.91 pp	—
	s1s2_3	-0.11%	+0.00%	+0.12 pp	—
	Point	Measured	Predicted	Error	Δt

Single-Issue Floating-Point Validation

The prior holdout set was entirely integer. Dhrystone and CoreMark contain no dynamic FLOAT instructions in the measured windows, so the FPU path had not been validated. I added `benchmarks/fpbench/`, a self-contained dense double matrix kernel, instead of importing Whetstone. The reason was practical: `fpbench` uses no `libm`, has no host-dependent timing harness, and keeps FP work from being constant-folded with volatile matrices and a checksum.

Two independent baseline simulator runs both measured 1,646,171 cycles and 739,818 retired instructions for `ITERATIONS=100`. The measured window's decoded mix is:

FU class	Instructions	Share
ALU	354,709	47.945%
FLOAT	211,400	28.575%
MEMORY	115,204	15.572%
BRANCH	58,500	7.907%

The FLOAT sub-ops are 211,300 FMA-path operations (`fmadd`, `fadd`, `fsub`, `fmul`) and 100 integer-to-float conversions, so the benchmark exercises the normal FP datapath rather than accidentally compiling to soft-float.

FPU timing. The single-issue makefile defines `spfp` and `dpfp`, but not `hardfloat`; `src/fpu/fpu.bsv` therefore instantiates `fpu_bsvfloat`. The model now detects this from `makefile.inc`. For a `hardfloat` build it reads `fma_stages`, `ftoi_stages`, `itof_stages`, and `ftof_stages` from `src/fpu/fpu.defines`. For the active `bsv_float` build, the FMA path is five cycles: one top-level input FIFO cycle plus the four explicit `fpu_fm_add_sub` state-machine stages before the result is visible. The FPU ready signal lags the result by one cycle because it observes `rg_multicycle_op` and `ff_input.notEmpty`.

Model	Cycles	IPC	Cycle error	Δt accuracy
Before FP latency	812,847	0.9102	-50.622%	64.612%
After FP latency	1,645,141	0.4497	-0.063%	99.863%

The integer baselines do not regress: Dhrystone remains 99.655% Δt accurate with -0.446% cycle error, and CoreMark remains 99.150% Δt accurate with -0.790% cycle error.

FP buffer-depth ladder. The integer-only conclusion that `isb_s3s4` depths 2, 3 and 4 are all free is too broad. Under FP load, depth 2 costs cycles:

Point	RTL delta	Model delta	Error	Δt accuracy
<code>s3s4_1</code>	+42.554%	+45.310%	+2.756 pp	92.118%
<code>s3s4_2</code>	+3.159%	+3.118%	-0.041 pp	99.768%
<code>s3s4_3</code>	+0.000%	+0.000%	+0.000 pp	99.863%
<code>s3s4_4</code>	+0.000%	+0.000%	+0.000 pp	99.863%
<code>s4s5_1</code>	+21.783%	+21.852%	+0.069 pp	99.985%
<code>s4s5_2</code>	+0.000%	+0.000%	+0.000 pp	99.863%

So the area conclusion is revised: `s3s4=2` is not free for FP throughput, while `s3s4=3`, `s3s4=4`, and `s4s5=2` still match the capacity-8 baseline on this workload.

Dual-issue FLOAT pairing. The dual-issue source was checked read-only. `stage2.bsv` allows `ALU+FLOAT`, `FLOAT+ALU`, `CONTROL+FLOAT`, and `FLOAT+CONTROL`, where `CONTROL` is `BRANCH`, `JAL`, or `JALR`. `FLOAT+FLOAT` and unlisted scarce-FU combinations fall through to single issue. The model's `FLOAT` pairing whitelist matches the RTL source, but there is still no dynamic dual-issue FP validation trace.

Calibration Log

Change	Effect	Evidence
Dual trace instrumentation	Added a testbench-only cycle counter to both dual commit-log slots; no core datapath files were changed.	The stamped CoreMark run reproduced the clean dual app-log counters exactly: 10,398,758 cycles and 12,771,408 instructions.
Multi-line <code>app_log</code> parsing	Fixed parser support for SHAKTI logs that print <code>IPC_MEASURE cycles</code> and <code>IPC_MEASURE instret</code> on separate lines, including the space before the colon.	The selected benchmark window now has 12,771,408 instructions, matching CoreMark ground truth.

Change	Effect	Evidence
WAW/WAR pair handling	Kept WAW and WAR co-issue enabled because <code>no_wawstalls</code> compiles out the checks.	Unit tests verify same-rd and write-after-read pairs commit together.
Lockstep scoping	Kept lockstep only for real paired bundles; a single bundle does not wait on slot 1.	Unit tests exercise dependent and independent single/pair cases.
Dual bypass shape	Fixed bypass lookup to inspect slot 1 and slot 0 of both downstream sources, matching <code>bypass.bsv</code> priority <code>src0/slot1 > src0/slot0 > src1/slot1 > src1/slot0</code> .	Full CoreMark dual baseline moved from IPC 1.0357 before this fix to IPC 1.2060.
Memory pairing policy	Split the actual delivered branch (<code>memory_pairing=none</code>) from the proposed second-port experiment (<code>memory_pairing=all</code>).	Run A is now a true second-memory-port experiment instead of the narrower, gated store-involving <code>dual_mem</code> path.
Guarded FIFO backpressure	Modelled <code>mkSizedFIFO</code> as a guarded FIFO where <code>FULL_N</code> is based on cycle-start occupancy, while <code>mkLFIFO</code> remains loopy. Added split compressed fetch-word handling for the single-issue front end.	Dhrystone <code>s3s4_1</code> moved from a near-zero prediction to 322,593 model cycles versus 321,804 RTL; CoreMark <code>s0s1_1</code> is now predicted within 1% aggregate.
CoreMark holdout windowing	Fixed <code>holdout/predict.py</code> so CoreMark variants use the baseline app-log window when only the baseline CoreMark trace is archived.	Removed false instruction-stream drift warnings and replaced invalid variant Δt values with aggregate-only CoreMark validation.
Floating-point timing	Added FP operation classes to the decoder and FPU latency parameters to <code>Model</code> . The active single-issue build uses <code>bsv_float</code> , not <code>hardfloat</code> , so FMA latency is source-derived from the <code>fpu_fm_add_sub</code> state machine rather than from a fitted constant.	<code>fpbench</code> accuracy moved from 64.612% to 99.863% Δt , and the integer baselines did not move.

Analysis

Run A, the second memory port, improves IPC by +0.698%. It removes the modeled non-RAW mem+mem reject counter, but the speedup is much smaller than a first-order lost-slot estimate. The measured interaction is sub-additive, not lockstep-gated: A+B is smaller than A plus B, so those changes remove overlapping bubbles rather than unlocking one another.

Run B improves IPC by +0.689%, confirming that stage3/stage4 coupling is a real limiter. Run C is sub-additive relative to A+B, so the second memory port and decoupling overlap in the bubbles they remove.

Run E is the largest individual improvement at +6.237%. It directly attacks the RAW rejection class and roughly halves the modeled RAW reject count. This is the strongest candidate for RTL work if its implementation cost is manageable.

Run H gives only +0.480%. The model’s independent branch+branch opportunity count is 159,885 on CoreMark, so branch+branch is not the dominant issue in this trace. This is also a useful independent replacement for the defective RTL event 52.

The automatic stacked run I reached IPC 1.2756. The extra probes show why it is not the actual best: A+E reached IPC 1.2619, while E+H reached IPC 1.2953. So the second memory port interferes with the intra-forwarding win in this trace, whereas branch+branch is mildly complementary with intra-forwarding.

Sensitivity

Config	Cycles	IPC	Δ IPC
instr_queue=2	14,681,039	0.8699	-27.868%
instr_queue=4	10,572,287	1.2080	+0.165%
instr_queue=6	10,589,781	1.2060	+0.000%
instr_queue=8	10,590,228	1.2060	-0.004%
instr_queue=12	10,590,778	1.2059	-0.009%
mispredict_penalty=0	10,372,390	1.2313	+2.096%
mispredict_penalty=1	10,589,781	1.2060	+0.000%
mispredict_penalty=2	10,807,038	1.1818	-2.010%
mispredict_penalty=3	11,024,295	1.1585	-3.941%
memory_latency=0	10,589,781	1.2060	+0.000%
memory_latency=1	12,222,907	1.0449	-13.361%
memory_latency=2	14,185,349	0.9003	-25.347%
memory_latency=3	16,155,195	0.7905	-34.450%

What Did Not Work

- Independent retire alone had no effect because actual lockstep stage4 already delivers paired slots to stage5 together.
- The branch next-PC relaxation is unresolved: the named F config remains an exact no-op in this model and should not be treated as RTL evidence.
- Symmetric slots remains an exact no-op in this trace/model. Treat it as an unexpressed hypothesis, not proof that the RTL change is worthless.
- Combining decoupled lockstep with intra-bundle forwarding did not drain on a 100k-instruction diagnostic run. The final best combination therefore excludes B when E is selected; this is reported as a model limitation/negative interaction, not hidden.
- Quad issue did not complete/drain in the full-trace sweep. It is reported as not completed rather than as a measured null.

Experiment Validation Coverage

Experiment	Predicted gain	Validation status	Notes
Baseline	+0.000%	model-only	No specific validation mapping recorded.
A: second memory port	+0.698%	below/near floor	No direct dual memory-port RTL point; fixed memory-latency sensitivity makes the knob active.
B: decouple lockstep	+0.689%	below/near floor	Coupling logic is model-only; effect is near the CoreMark small-effect floor.
C: A+B	+1.015%	near floor	Sub-additive; above small-effect floors but below overall mechanism floor.
D: independent retire	+0.000%	below resolution	Implemented flag has exact zero effect because upstream delivery remains paired.
E: intra-bundle ALU forwarding	+6.237%	no ground truth	Bypass/forwarding has no buildable single-issue design point.
F: relax branch next-PC stall	+0.000%	unimplemented/no-op	Exact zero; do not report as an RTL null.
G: symmetric slots	+0.000%	unimplemented/no-op	Exact zero in this trace/model; do not report as an RTL null.

Experiment	Predicted gain	Validation status	Notes
H: second branch unit	+0.480%	near/below floor	Small model-owned branch-pair opportunity result.
I: best combination	+5.772%	partial	Two-issue stack; inherits E's no-ground-truth forwarding limitation.
J: quad issue on best	n/a	not completed	Full-trace quad approximation did not drain/complete.
A+E: memory + intra forwarding	+4.635%	partial	Shows memory port interferes with E.
E+H: intra forwarding + branch	+7.406%	partial	Best draining two-issue model result; inherits E limitation.

Small-effect floors are 0.17 pp on Dhrystone and 0.64 pp on CoreMark; claims at or below that scale should be read as below model resolution.

Recommendations

1. Prototype intra-bundle ALU forwarding first. Estimated standalone gain: +6.237%; rough RTL cost: medium (operand muxing, bypass select, verification of RAW edge cases).
2. Combine intra-bundle ALU forwarding with a second branch unit only if the branch hardware is cheap. Best measured combination: E+H at IPC 1.2953.
3. Treat the second memory port as a secondary change, not the headline by itself. Estimated standalone gain: +0.698%; rough RTL cost: high (D-cache/store-buffer/arbitrator integration). It regresses the E combination, so do not prioritize it before validating with a dual trace.
4. Investigate lockstep decoupling separately from intra-forwarding. Estimated standalone gain: +0.689%; rough RTL cost: medium-high because it touches stage3/stage4 valid/ready and commit bookkeeping.
5. Deprioritize branch+branch hardware until a corrected RTL counter or dual trace says otherwise. Estimated gain: +0.480%; rough RTL cost: medium.
6. Do not commit to quad issue from this evidence. The current approximation did not drain on the full trace and is not a validated 4-wide front end.

Limitations

The model is a timing model over an already committed dynamic trace. It does not compute data values. The validation set now covers small-footprint, cache-resident integer code plus one dense floating-point arithmetic kernel. It still does not model data-cache misses, cache conflicts, instruction-cache misses, wrong-path fetch bandwidth, FP divide/sqrt data-dependent special cases, or CSR/trap behavior that depends on values. The dual CoreMark trace now provides per-instruction RTL commit cycles, but the baseline model only reaches the accuracy reported above; the residual mismatches are concentrated around multiply-heavy and control-flow neighborhoods and should be diagnosed before treating the experiment deltas as RTL-commitment evidence. A Dhrystone commit trace is still missing for held-out dual-issue Δt validation, and no dynamic dual-issue FP trace exists yet.